Juwen Lie

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PROFESSIONAL SUMMARY

Mechanical DVT and NPI engineer for consumer electronics (IP cameras, CCTV, Wi-Fi), with an aerospace engineering background. Product-minded and systems-driven: leads cross-functional teams and root-cause analysis, owns programs from EVT to mass production, and builds software and AI systems on the side.

EXPERIENCE

R&D; Mechanical Engineer

PEGATRON / PT Pegatron Technology Indonesia — Batam, Indonesia | Nov 2023 – Present

- Lead mechanical DVT for IPX6/IPX7 IP camera and CCTV programs — water leakage, focus/SFR, lens-holder, antenna and cable routing, acoustics, and spring/pin mechanisms — from EVT through mass production.
- Manage 6–12 month EVT-to-MP programs and 2–3 concurrent Wi-Fi projects (1–3 SKUs each), keeping schedule, quality and cost on track.
- Lead root-cause analysis of production-impacting issues using measurement, tolerance stack-up and DOE-style comparison; define short- and long-term fixes.
- Coordinate PM, ME, EE, RF, quality, factory and suppliers; lead customer talks and plan rework/redesign.
- Design in SolidWorks (3D/2D); build 3D-printed mockups, verify assembly, and author assembly SOPs.
- Manage plastic/metal parts, shielding, heat spreaders, gaskets, Mylar and die-cut liners; run HSF/material approval and second-source activities.
- Support QSA/QPA and ISO 9001 supplier audits; prepare FA reports and technical recommendations.
- Lead a team of 2 engineers; mentored 5–7 in NPI, failure analysis, measurement and part approval.
- International R&D; assignment at PEGATRON, Suzhou, China.

Mechanical & Fabrication Engineer

PT Anugerah Tirta Indonesia — Indonesia | Aug 2023 – Oct 2023

- Coordinated water-tank projects, schedules, BOM and stock across Indonesia, plus pipe installation for new housing.
- Prepared Bill of Quantity cost calculations, trash-rack cutting plans, and installation-cost analysis.
- Coordinated Chinese-language communication between China-based vendors and Indonesian project teams.

SELECTED PROJECTS

IP Camera & CCTV — Mechanical DVT — Mechanical | Reliability | NPI — reliability validation (waterproofing, focus/SFR, acoustics, mechanisms) across EVT→MP.

Orbis — Software | Python, SQLite, PWA — zero-dependency backend + offline-first PWA for communication, identity and growth tracking.

Peers — Software | PWA — offline-first single-file app for verified reviews of food, travel and products.

House of Articles — Product | Data modelling — athlete-moment sneaker news + live market feed on a versioned JSON data spine.

EDUCATION

Tamkang University — Taiwan

Bachelor's Degree, Aerospace, Aeronautical and Astronautical Engineering | 2018 – 2022

- Academic honors; GPA 3.847/4.0 (Rank 5 of 76).
- Research: preconditioning method for 1D Euler equations on a diverging nozzle (vs Roe, Van Leer, Steger-Warming); research proposal on numerical simulation of two-phase flow in geothermal wells.

SKILLS

Mechanical: SolidWorks (3D/2D), AutoCAD, tolerance stack-up, plastic injection, thermal

Validation: Failure analysis, root-cause analysis, DOE-style comparison, CMM/OMM measurement
Process: BOM, PLM/SRM/ECN, RFQ & supplier coordination, part approval, QSA/QPA, ISO 9001
Software & Tech: Python, HTML/CSS/JavaScript, SQLite, systems design
Languages: English (IELTS 7.5), Indonesian (native), Mandarin (working proficiency)